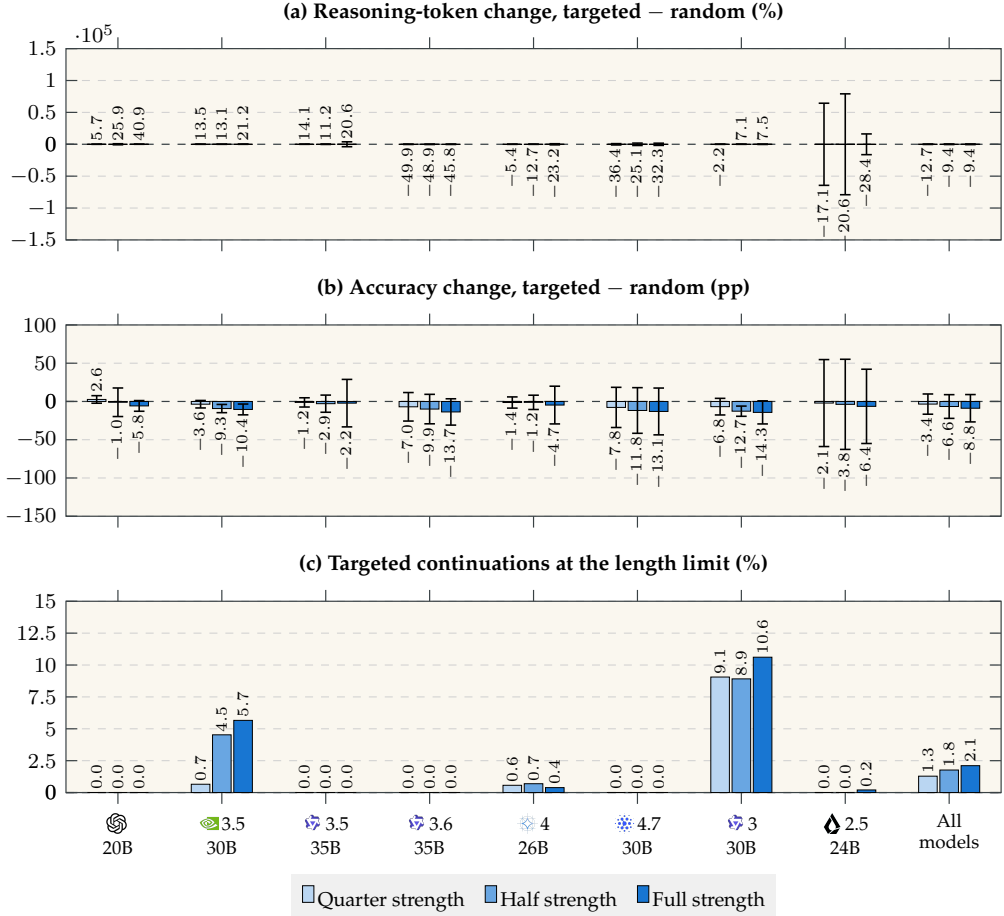

(a) Reasoning-token change, targeted − random (%)
·10^5
1.5
1
0.5
0
−0.5
−1
−1.5
5.7
25.9
40.9
13.5
13.1
21.2
14.1
11.2
20.6
−49.9
−48.9
−45.8
−5.4
−12.7
−23.2
−36.4
−25.1
−32.3
−2.2
7.1
7.5
−17.1
20.6
−28.4
−12.7
−9.4
−9.4
(b) Accuracy change, targeted − random (pp)
100
50
0
−50
−100
−150
2.6
−1.0
−5.8
−3.6
−9.3
−10.4
−1.2
−2.9
−2.2
−7.0
−9.9
−13.7
−1.4
−1.2
−4.7
−7.8
−11.8
−13.1
−6.8
−12.7
−14.3
−2.1
−3.8
−6.4
−3.4
−6.6
−8.8
(c) Targeted continuations at the length limit (%)
15
12.5
10
7.5
5
2.5
0
0.0
0.0
0.0
0.7
4.5
5.7
0.0
0.0
0.0
0.0
0.0
0.0
0.6
0.7
0.4
0.0
0.0
0.0
9.1
8.9
10.6
0.0
0.0
0.2
1.3
1.8
2.1
20B
3.5 30B
3.5 35B
3.6 35B
4 26B
4.7 30B
3 30B
2.5 24B
All models
Quarter strength
Half strength
Full strength